



Ph.D. Positions in Trustworthy and Multi-Agent GeoAI

Group of Dr. Peng Luo · Incoming Assistant Professor, August 2026

School of Earth, Environment, and Sustainability · University of Iowa

I will join the School of Earth, Environment, and Sustainability (SEES) at the University of Iowa as a tenure-track Assistant Professor in August 2026, and I am building a new research group at the intersection of Geographic Information Science (GIScience), Geospatial Artificial Intelligence (GeoAI), and spatial data science. I am recruiting two fully funded Ph.D. students to begin in Fall 2027. Each position is fully funded through a research or teaching assistantship, covering a stipend and tuition.

Research Directions

Position 1 - Trustworthy GeoAI. Developing reliable, interpretable, and uncertainty-aware geospatial AI. Topics include spatial explicit modeling, geospatial explainable AI (XAI), and uncertainty quantification of GeoAI models for spatial prediction and association analysis.

Position 2 - Multi-Agent GeoAI Modeling. Building LLM-driven, multi-agent frameworks for automated geospatial model discovery, reasoning, and scalable, self-evolving spatial analysis.

Qualifications

- A Bachelor's or Master's degree in GIScience, computer science, statistics, remote sensing, geography, or a closely related field.
- Strong programming skills (e.g., Python) and a genuine interest in spatial data science and AI.
- Self-motivated, curious, and able to communicate clearly in written and spoken English. Prior research experience is a plus but is not required.

What I Offer

- Full funding through a research or teaching assistantship, covering a stipend and tuition for the duration of the Ph.D.
- Close, individualized mentorship and a clear path toward an academic or research career.
- An active international collaboration network and support for conference travel, computing resources, and publications.

Mentoring Philosophy

My mentoring philosophy is grounded in a fundamental Kantian principle: always treat people as ends in themselves, never merely as means. Guided by this conviction, my hopes and expectations for our partnership follow a clear hierarchy.

First - self-discovery over output.

My foremost responsibility is not to maximize your external achievements or output, but to safeguard the space and opportunity for you to fully experience the doctoral journey. I have walked that road myself; I tasted both its sweetness and its hardship. Yet in the end, looking back, I realized that my greatest reward was a deeper understanding of who I am - and for that, I remain profoundly grateful for the journey. A

Ph.D. is, first and foremost, a process of coming to know yourself on your own terms.

Second - intellectual freedom and good taste.

Building on that, I value your intellectual freedom above all. I hope you will find the geographic questions that truly interest you, and come to believe that you are the right person to work on them. In the age of AI, I think a researcher's taste matters more than ever. Publishing many average papers is not the goal - one piece of work with real taste is worth far more. And good taste only comes when you genuinely care about a problem: you do your best work when you are doing what you love.

Third - strategic alignment and shared growth.

To be entirely transparent: as a junior faculty member, I navigate the real pressures of academic survival and promotion. While the first two principles always take precedence, I hope your interests can intersect with my own research trajectory. When our paths align, I can offer my most expert guidance with full confidence, and we both thrive.

How to Reach Me

If you are interested, please email me directly at peng-luo@uiowa.edu with your CV, transcripts, and a brief statement of research interests (indicate which position interests you), using the subject line "Prospective Ph.D. - [your name]." I review inquiries on a rolling basis and will follow up with promising candidates for a conversation. I encourage reaching out before submitting a formal application.

Formal application: Admitted students enroll through the Ph.D. in Geography program (GIScience track) at SEES. The University's application deadline is December 1 for the following Fall. I will guide selected candidates through the process - please contact me first.

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